



DASVIR 30/60 **(DACLATASVIR TABLETS 30 mg & 60 mg)**

Name and strength of Active Ingredient

Dasvir 30mg

Each film coated Tablet contains Daclatasvir dihydrochloride equivalent to Daclatasvir 30mg

Dasvir 60 mg

Each film coated Tablet contains Daclatasvir dihydrochloride equivalent to Daclatasvir 60mg

Product Description

Dasvir 30mg

Yellow round; bevel edged, biconvex, film coated tablets debossed with 'D 14' on one side and 'H' on the other side.

Dasvir 60 mg

Light yellow, round, bevel edged, biconvex film coated tablets debossed with 'D 19' on one side and 'H' on other side.

Dosage form

Film coated Tablets

Pharmacodynamics

Pharmacotherapeutic group: Direct-acting antiviral ATC code, J05AP07

Mechanism of action

Daclatasvir is an inhibitor of nonstructural protein 5A (NS5A), a multifunctional protein that is an essential component of the HCV replication complex. Daclatasvir inhibits both viral RNA replication and virion assembly.

Antiviral activity in cell culture

Daclatasvir is an inhibitor of HCV genotypes 1a and 1b replication in cell-based replicon assays with effective concentration (50% reduction, EC₅₀) values of 0.003-0.050 and 0.001-0.009 nM, respectively, depending on the assay method. The daclatasvir EC₅₀ values in the replicon system were 0.003-1.25 nM for genotypes 3a, 4a, 5a and 6a, and 0.034-19 nM for genotype 2a as well as 0.020 nM for infectious genotype 2a (JFH-1) virus.

Daclatasvir showed additive to synergistic interactions with interferon alfa, HCV nonstructural protein 3 (NS3) PIs, HCV nonstructural protein 5B (NS5B) non-nucleoside inhibitors, and HCV NS5B nucleoside analogues in combination studies using the cell-based HCV replicon system. No antagonism of antiviral activity was observed.

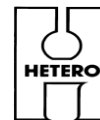
No clinically relevant antiviral activity was observed against a variety of RNA and DNA viruses, including HIV, confirming that daclatasvir, which inhibits a HCV-specific target, is highly selective for HCV.

Resistance in cell culture

Substitutions conferring daclatasvir resistance in genotypes 1-4 were observed in the N-terminal 100 amino acid region of NS5A in a cell-based replicon system. L31V and Y93H were frequently observed resistance substitutions in genotype 1b, while M28T, L31V/M, Q30E/H/R, and Y93C/H/N were frequently observed resistance substitutions in genotype 1a. These substitutions conferred low level resistance (EC₅₀<1 nM) for genotype 1b, and higher levels of resistance for genotype 1a (EC₅₀ up to 350 nM). The most resistant variants with single amino acid substitution in genotype 2a and genotype 3a were F28S (EC₅₀>300 nM) and Y93H (EC₅₀>1,000 nM), respectively. In genotype 4, amino acid substitutions at 30 and 93 (EC₅₀< 16 nM) were frequently selected.

Cross-resistance

HCV replicons expressing daclatasvir-associated resistance substitutions remained fully sensitive to interferon alfa and other anti-HCV agents with different mechanisms of action, such as NS3 protease and NS5B polymerase (nucleoside and non-nucleoside) inhibitors.



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Clinical efficacy and safety

In clinical studies of daclatasvir in combination with sofosbuvir or with peginterferon alfa and ribavirin, plasma HCV RNA values were measured using the COBAS TaqMan HCV test (version 2.0), for use with the High Pure System, with a lower limit of quantification (LLOQ) of 25 IU/ ml. SVR was the primary endpoint to determine the HCV cure rate, which was defined as HCV RNA less than LLOQ at 12 weeks after the end of treatment (SVR12) for AI444040, ALLY-1 (AI444215), ALLY-2 (AI444216), ALLY-3 (AI444218), AI444042 and as HCV RNA undetectable at 24 weeks after the end of treatment (SVR24) for study AI444010.

Daclatasvir in combination with sofosbuvir

The efficacy and safety of daclatasvir 60 mg once daily in combination with sofosbuvir 400 mg once daily, with or without ribavirin, in the treatment of infection with chronic HCV genotype 1, 2, or 3 were evaluated in four open-label studies (AI444040, ALLY-1, ALLY-2 and ALLY-3).

In study AI444040, 211 adults with HCV genotype 1, 2, or 3 infection and without cirrhosis received daclatasvir and sofosbuvir, with or without ribavirin. Among the 167 patients with HCV genotype 1 infection, 126 were treatment-naïve and 41 had failed prior therapy with a PI regimen (boceprevir or telaprevir). All 44 patients with HCV genotype 2 (n=26) or 3 (n=18) infection were treatment-naïve. Treatment duration was 12 weeks for 82 treatment-naïve HCV genotype 1 patients, and 24 weeks for all other patients in the study. The 211 patients had a median age of 54 years (range: 20 to 70); 83% were white; 12% were black/African-American; 2% were Asian; 20% were Hispanic or Latino. The mean score on the FibroTest (a validated non-invasive diagnostic assay) was 0.460 (range: 0.03 to 0.89). Conversion of the FibroTest score to the corresponding METAVIR score suggests that 35% of all patients (49% of patients with prior PI failure, 30% of patients with genotype 2 or 3) had F3 liver fibrosis. Most patients (71%, including 98% of prior PI failures) had IL-28B rs12979860 non-CC genotypes.

SVR12 was achieved by 99% patients with HCV genotype 1, 96% of those with genotype 2 and 89% of those with genotype 3 (see Tables 1 and 2). Response was rapid (viral load at Week 4 showed that more than 97% of patients responded to therapy) and was not influenced by HCV subtype (1a/1b), IL28B genotype, or use of ribavirin. Among treatment-naïve patients with HCV RNA results at both follow-up Weeks 12 and 24, concordance between SVR12 and SVR24 was 99.5% independent of treatment duration.

Treatment-naïve patients with HCV genotype 1 who received 12 weeks of treatment had a similar response as those treated for 24 weeks (Table 1).

Table 1: Treatment outcomes, daclatasvir in combination with sofosbuvir, HCV genotype 1 in Study AI444040

	Treatment naïve			Prior telaprevir or boceprevir failures		
	Daclatasvir + Sofosbuvir N=70	Daclatasvir + Sofosbuvir + Ribavirin N=56	All N=126	Daclatasvir + Sofosbuvir N=21	Daclatasvir + Sofosbuvir + Ribavirin N=20	All N=41
End of treatment HCV RNA undetectable	70 (100%)	56 (100%)	126 (100%)	19 (91%)	19 (95%)	38 (93%)
SVR12 (overall)*	70 (100%)	55 (98%) *	125 (99%) *	21 (100%)	20 (100%)	41 (100%)
12 weeks treatment duration	41/41 (100%)	40/41 (98%)	81/82 (99%)	-	-	-
24 weeks treatment duration	29/29 (100%)	15/15 (100%)	44/44 (100%)	21 (100%)	20 (100%)	41 (100%)
≥ F3 Liver Fibrosis	-	-	41/41 (100%)	-	-	20/20 (100%)

*Patients who had missing data at follow-up Week 12 were considered responders if their next available HCV RNA value was <LLOQ. One treatment-naïve patient was missing both post-treatment Weeks 12 and 24 data.

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Table 2: Treatment outcomes, daclatasvirin combination with sofosbuvir for 24 weeks, treatment naive patients with HCV genotype 2 or 3 in Study AI444040

	Genotype 2			Genotype 3		
	Daclatasvir + Sofosbuvir N=17	Daclatasvir + Sofosbuvir + Ribavirin N=9	All Genotype 2 N=26	Daclatasvir + Sofosbuvir N=13	Daclatasvir + Sofosbuvir + Ribavirin N=5	All Genotype 3 N=18
End of treatment HCV RNA undetectable	17 (100%)	9 (100%)	26 (100%)	11 (85%)	5 (100%)	16 (89%)
SVR12 *	17 (100%)	8 (89%) *	25 (96%) *	11 (85%)	5 (100%)	16 (89%)
≥ F3 Liver Fibrosis			8/8 (100%)			5/5 (100%)
Virologic failure						
Virologic breakthrough**	0	0	0	1 (8%)	0	1 (6%)
Relapse**	0	0	0	1/11 (9%)	0	1/16 (6%)

* Patients who had missing data at follow-up Week 12 were considered responders if their next available HCV RNA value was <LLOQ. One patient with HCV genotype 2 infection was missing both post- treatment Week 12 and 24 data.

** The patient with virologic breakthrough met the original protocol definition of confirmed HCV RNA <LLOQ, detectable at treatment Week 8. Relapse was defined as HCV RNA ≥ LLOQ during follow-up after HCV RNA <LLOQ at end of treatment. Relapse includes observations through follow-up Week 24.

Advanced cirrhosis and post-liver transplant (ALLY-1)

In study ALLY-1, the regimen of daclatasvir, sofosbuvir, and ribavirin administered for 12 weeks was evaluated in 113 adults with chronic hepatitis C and Child-Pugh A, B or C cirrhosis (n=60) or HCV recurrence after liver transplantation (n=53). Patients with HCV genotype 1, 2, 3, 4, 5 or 6 infection were eligible to enroll. Patients received daclatasvir 60 mg once daily, sofosbuvir 400 mg once daily, and ribavirin (600 mg starting dose) for 12 weeks and were monitored for 24 weeks post treatment. Patients demographics and main disease characteristics are summarised in Table 3.

Table 3: Demographics and main disease characteristics in Study ALLY-1

	Cirrhotic cohort	Post-Liver Transplant
	N = 60	N = 53
Age (years): median (range)	58 (19-75)	59 (22-82)
Race:		
White	57 (95%)	51 (96%)
Black/African American	3 (5%)	1 (2%)
Other	0	1 (2%)
HCV genotype:		
1a	34 (57%)	31 (58%)
1b	11 (18%)	10 (19%)
2	5 (8%)	0
3	6 (10%)	11 (21%)
4	4 (7%)	0

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6	0	1 (2%)
Fibrosis stage		
F0	0	6 (11%)
F1	1 (2%)	10 (19%)
F2	3 (5%)	7 (13%)
F3	8 (13%)	13 (25%)
F4	48 (80%)	16 (30%)
Not reported	0	1 (2%)
CP classes		ND
CP A	12 (20%)	
CP B	32 (53%)	
CP C	16 (27%)	
MELD score		ND
mean	13.3	
median	13.0	
Q1, Q3	10, 16	
Min, Max	8, 27	

ND: Not determined

SVR12 was achieved by 83% (50/60) of patients in the cirrhosis cohort, with a marked difference between patients with Child-Pugh A or B (92-94%) as compared to those with Child-Pugh C and 94% of patients in the post-liver transplant cohort (Table 4). SVR rates were comparable regardless of age, race, gender, IL28B allele status, or baseline HCV RNA level. In the cirrhosis cohort, 4 patients with hepatocellular carcinoma underwent liver transplantation after 1 -71 days of treatment; 3 of the 4 patients received 12 weeks of post-liver transplant treatment extension and 1 patient, treated for 23 days before transplantation, did not receive treatment extension. All 4 patients achieved SVR12.

Table 4: Treatment outcomes, daclatasvir in combination with sofosbuvir and ribavirin for 12 weeks, patients with cirrhosis or HCV recurrence after liver transplantation, Study ALLY-1

	Cirrhotic cohort		Post-Liver Transplant	
	N=60		N=53	
End of treatment HCV RNA undetectable	58/60 (97%)		53/53 (100%)	
	SVR12	Relapse	SVR12	Relapse
All patients	50/60 (83%)	9/58* (16%)	50/53 (94%)	3/53 (6%)
Cirrhosis			ND	ND
CP A	11/12 (92%)	1/12 (8%)		
CP B	30/32 (94%)	2/32 (6%)		
CP C	9/16 (56%)	6/14 (43%)		
Genotype 1	37/45 (82%)	7/45 (16%)	39/41 (97%)	2/41 (5%)
1a	26/34 (77%)	7/33 (21%)	30/31 (97%)	1/31 (3%)
1b	11/11 (100%)	0%	9/10 (90%)	1/10 (10%)

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Genotype 2	4/5 (80%)	1/5 (20%)	--	--
Genotype 3	5/6 (83%)	1/6 (17%)	10/11 (91%)	1/11 (9%)
Genotype 4	4/4 (100%)	0%	--	--
Genotype 6	--	--	1/1 (100%)	0%

ND: Not determined

* 2 patients had detectable HCV RNA at end of treatment; 1 of these patients achieved SVR.

HCV/HIV co-infection (ALLY-2)

In study ALLY-2, the combination of daclatasvir and sofosbuvir administered for 12 weeks was evaluated in 153 adults with chronic hepatitis C and HIV co-infection; 101 patients were HCV treatment-naïve and 52 patients had failed prior HCV therapy. Patients with HCV genotype 1, 2, 3, 4, 5, or 6 infection were eligible to enroll, including patients with compensated cirrhosis (Child- Pugh A). The dose of daclatasvir was adjusted for concomitant antiretroviral use. Patient demographics and baseline disease characteristics are summarised in Table 5.

Table 5: Demographics and baseline characteristics in Study ALLY-2

Patient disposition	Daclatasvir + sofosbuvir 12 weeks N=153
Age (years): median (range)	53 (24-71)
Race:	
White	97 (63%)
Black/African American	50 (33%)
Other	6 (4%)
HCV genotype:	
1a	104 (68%)
1b	23 (15%)
2	13 (8%)
3	10 (7%)
4	3 (2%)
Compensated cirrhosis	24 (16%)
Concomitant HIV therapy:	
PI-based	70 (46%)
NNRTI-based	40 (26%)
Other	41 (27%)
None	2 (1%)

Overall, SVR12 was achieved by 97% (149/153) of patients administered daclatasvir and sofosbuvir for 12 weeks in ALLY-2. SVR rates were >94% across combination antiretroviral therapy (cART) regimens, including boosted-PI-, NNRTI-, and integrase inhibitor (INSTI)-based therapies. SVR rates were comparable regardless of HIV regimen, age, race, gender, IL28B allele status, or baseline HCV RNA level. Outcomes by prior treatment experience are presented in Table 6.

A third treatment group in study ALLY-2 included 50 HCV treatment-naïve HIV co-infected patients who received daclatasvir and sofosbuvir for 8 weeks. Demographic and baseline characteristics of these 50 patients were generally comparable to those for patients who received 12 weeks of study treatment. The SVR rate for patients treated for 8 weeks was lower with this treatment duration as summarized in Table 6.

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Table 6: Treatment outcomes, daclatasvirin combination with sofosbuvir in patients with HCV/HIV co-infection in Study ALLY-2

	8 weeks therapy	12 weeks therapy	
	HCV Treatment naive	HCV Treatment naive	HCV Treatment-experienced*
	N=50	N=101	N=52
End of treatment HCV RNA undetectable	50/50 (100%)	100/101 (99%)	52/52 (100%)
SVR12	38/50 (76%)	98/101 (97%)	51/52 (98%)
No cirrhosis**	34/44 (77%)	88/90 (98%)	34/34 (100%)
With cirrhosis**	3/5 (60%)	8/9 (89%)	14/15 (93%)
Genotype 1	31/41 (76%)	80/83 (96%)	43/44 (98%)
1a	28/35 (80%)	68/71 (96%)	32/33 (97%)
1b	3/6 (50%)	12/12 (100%)	11/11 (100%)
Genotype 2	5/6 (83%)	11/11 (100%)	2/2 (100%)
Genotype 3	2/3 (67%)	6/6 (100%)	4/4 (100%)
Genotype 4	0	1/1 (100%)	2/2 (100%)
Virologic failure			
Detectable at end of treatment	0	1/101 (1%)	0
Relapse	10/50 (20%)	1/101 (1%)	1/52 (2%)
Missing post-treatment data	2/50 (4%)	1/101 (1%)	0

* Mainly interferon-based therapy ± NS3/4 PI.

** Cirrhosis was determined by liver biopsy, FibroScan >14.6 kPa, or FibroTest score ≥0.75 aminotransferase (AST): platelet ratio index (APRI) >2. For 5 patients, cirrhosis status was indeterminate.

HCV Genotype 3 (ALLY-3)

In study ALLY-3, the combination of daclatasvir and sofosbuvir administered for 12 weeks was evaluated in 152 adults infected with HCV genotype 3; 101 patients were treatment-naïve and 51 patients had failed prior antiviral therapy. Median age was 55 years (range: 24 to 73); 90% of patients were white; 4% were black/African-American; 5% were Asian; 16% were Hispanic or Latino. The median viral load was 6.42 log₁₀ IU/ml, and 21% of patients had compensated cirrhosis. Most patients (61%) had IL-28B rs12979860 non-CC genotypes.

SVR12 was achieved in 90% of treatment-naïve patients and 86% of treatment-experienced patients. Response was rapid (viral load at Week 4 showed that more than 95% of patients responded to therapy) and was not influenced by IL28B genotype. SVR12 rates were lower among patients with cirrhosis (see Table 7).

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Table 7: Treatment outcomes, daclatasvirin combination with sofosbuvir for 12 weeks, patients with HCV genotype 3 in Study ALLY-3

	Treatment-naïve N=101	Treatment-experienced* N=51	Total N=152
End of treatment HCV RNA undetectable	100 (99%)	51 (100%)	151 (99%)
SVR12	91 (90%)	44 (86%)	135 (89%)
No cirrhosis[§]	73/75 (97%)	32/34 (94%)	105/109 (96%)
With cirrhosis[§]	11/19 (58%)	9/13 (69%)	20/32 (63%)
Virologic failure			
Virologic breakthrough	0	0	0
Detectable HCV RNA at end of treatment	1 (1%)	0	1 (0.7%)
Relapse	9/100 (9%)	7/51 (14%)	16/151 (11%)

* Mainly interferon-based therapy, but 7 patients received sofosbuvir + ribavirin and 2 patients received acyclophilin inhibitor.

[§] Cirrhosis was determined by liver biopsy (METAVIR F4) for 14 patients, FibroScan >14.6 kPa for 11 patients or FibroTest score ≥0.75 and aspartate aminotransferase (AST): platelet ratio index (APRI) >2 for 7 patients. For 11 patients, cirrhosis status was missing or inconclusive (FibroTest score >0.48 to <0.75 or APRI >1 to ≤2).

Daclatasvir in combination with peginterferon alfa and ribavirin

AI444042 and AI444010 were randomised, double-blind studies that evaluated the efficacy and safety of daclatasvir in combination with peginterferon alfa and ribavirin (pegIFN/RBV) in the treatment of chronic HCV infection in treatment-naïve adults with compensated liver disease (including cirrhosis). AI444042 enrolled patients with HCV genotype 4 infection and AI444010 enrolled patients with either genotype 1 or 4.

AI444042: Patients received daclatasvir 60 mg once daily (n=82) or placebo (n=42) plus pegIFN/RBV for 24 weeks. Patients in the daclatasvir treatment group who did not have HCV RNA undetectable at both Weeks 4 and 12 and all placebo-treated patients continued pegIFN/RBV for another 24 weeks. Treated patients had a median age of 49 years (range: 20 to 71); 77% of patients were white; 19% were black/African-American; 4% were Hispanic or Latino. Ten percent of patients had compensated cirrhosis, and 75% of patients had IL-28B rs12979860 non-CC genotypes. Treatment outcomes in study AI444042 are presented in Table 13. Response was rapid (at Week 4 91% of daclatasvir-treated patients had HCV RNA <LLOQ). SVR12 rates were higher for patients with the IL-28B CC genotype than for those with non-CC genotypes and for patients with baseline HCV RNA less than 800,000 IU/ml but consistently higher in the daclatasvir-treated patients than for placebo-treated patients in all subgroups.

AI444010: Patients received daclatasvir 60 mg once daily (n=158) or placebo (n=78) plus pegIFN/RBV through Week 12. Patients assigned to daclatasvir 60 mg once-daily treatment group who had HCV RNA <LLOQ at Week 4 and undetectable at Week 10 were then randomised to receive another 12 weeks of daclatasvir 60 mg +pegIFN/RBV or placebo +pegIFN/RBV for a total treatment duration of 24 weeks. Patients originally assigned to placebo and those in the daclatasvir group who did not achieve HCV RNA <LLOQ at Week 4 and undetectable at Week 10 continued pegIFN/RBV to complete 48 weeks of treatment. Treated patients had a median age of 50 years (range: 18 to 67); 79% of patients were white; 13% were black/African-American; 1% were Asian; 9% were Hispanic or Latino. Seven percent of patients had compensated cirrhosis; 92% had HCV genotype 1 (72% 1a and 20% 1b) and 8% had HCV genotype 4; 65% of patients had IL-28B rs12979860 non-CC genotypes.

Treatment outcomes in study AI444010 for patients with HCV genotype 4 are presented in Table 8. For HCV

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genotype 1, SVR12 rates were 64% (54% for 1a; 84% for 1b) for patients treated with daclatasvir + pegIFN/RBV and 36% for patients treated with placebo + pegIFN/RBV. For daclatasvir-treated patients with HCV RNA results at both follow-up Weeks 12 and 24, concordance of SVR12 and SVR24 was 97% for HCV genotype 1 and 100% for HCV genotype 4.

Table 8: Treatment outcomes, daclatasvir in combination with peginterferon alfa and ribavirin (pegIFN/RBV), treatment-naïve patients with HCV genotype 4

	Study AI444042		Study AI444010	
	daclatasvir + pegIFN/RBV	pegIFN/RB V	daclatasvir + pegIFN/RBV	pegIFN/RB V
	N=82	N=42	N=12	N=6
End of treatment HCV RNA undetectable	74 (90%)	27 (64%)	12 (100%)	4 (67%)
SVR12*	67 (82%)	18 (43%)	12 (100%)	3 (50%)
No cirrhosis	56/69 (81%)**	17/38 (45%)	12/12 (100%)	3/6 (50%)
With cirrhosis	7/9 (78%)**	1/4 (25%)	0	0
Virologic failure				
On-treatment virologic failure [§]	8 (10%)	15 (36%)	0	0
Relapse [§]	2/74 (3%)	8/27 (30%)	0	1/4 (25%)

* Patients who had missing data at follow-up Week 12 were considered responders if their next available HCV RNA value was <LLOQ.

** Cirrhosis status was not reported for four patients in the daclatasvir + pegIFN/RBV group

Long term efficacy data

Limited data are available from an ongoing follow-up study to assess durability of response up to 3 years after treatment with daclatasvir. Among patients who achieved SVR12 with daclatasvir and sofosbuvir (± ribavirin) with a median duration of post-SVR12 follow-up of 15 months, no relapses have occurred. Among patients who achieved SVR12 with daclatasvir + pegIFN/RBV with a median duration of post-SVR12 follow-up of 22 months, 1% of patients relapsed.

Resistance in clinical studies

Daclatasvir in combination with sofosbuvir

Frequency of baseline NS5A resistance-associated variants (RAVs)

Baseline NS5A RAVs were frequently observed in clinical studies of daclatasvir plus sofosbuvir +/- ribavirin, with approximately 11% in genotype 1 infection (GT1a: 28, 30, 31, or 93; GT1b: 31 or 93), 50% in genotype 2 infection (L31M), 8% in genotype 3 infection (Y93H) and in 71% in genotype 4 infection (L28M or L30R).

Impact of baseline NS5A RAVs on cure rates

The baseline NS5A RAVs described above had no major impact on cure rates in patients treated with sofosbuvir + daclatasvir +/- ribavirin, with the exception of the Y93H RAV in genotype 3 infection (seen in 16/192 [8%] of patients). The SVR12 rate in genotype-3 infected patients with this RAV is reduced (in practice as relapse after end of treatment response), especially in patients with cirrhosis. The overall cure rate for genotype-3 infected patients who were treated for 12 weeks with sofosbuvir + daclatasvir (without ribavirin) in the presence and absence of the Y93H RAV was 7/13 (54%) and 134/145 (92%), respectively. There was no Y93H RAV present at baseline for genotype-3 infected patients treated for 12-weeks with sofosbuvir + daclatasvir + ribavirin, and

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thus SVR outcomes cannot be assessed.

Emerging resistance

In a pooled analysis of 629 patients who received daclatasvir and sofosbuvir with or without ribavirin in Phase 2 and 3 studies for 12 or 24 weeks, 36 patients qualified for resistance analysis due to virologic failure or early study discontinuation and having HCV RNA greater than 1,000 IU/ml. Observed emergent NS5A resistance-associated variants are reported in Table 9.

Table 9: Summary of noted newly emergent HCV NS5A substitutions on treatment or during follow-up in treated non-SVR12 subjects infected with HCV genotypes 1 through 3

Category/ Substitution, n (%)	Genotype 1a N =301	Genotype 1b N=79	Genotype 2 N=44	Genotype 3 N=197
Non-responders (non-SVR12)	14*	1	2*	21**
with baseline and post- baseline sequence	12	1	1	20
with emergent NS5A RAVs***	10 (83%)	1 (100%)	0	16 (80%)
M28: T	2 (17%)	--	--	0
Q30: H, K, R	9 (75%)	--	--	--
L31: I, M, V	2 (17%)	0	0	1 (5%)
P32-deletion	0	1 (100%)	0	0
H58: D, P	2 (17%)	--	--	--
S62: L	--	--	--	2 (10%)
Y93: C, H, N	2 (17%)	0	0	11 (55%)

* One patient was lost to follow-up

** One patient considered a protocol failure (non-SVR) achieved SVR

*** NS5A RAVs monitored at amino acid positions are 28, 29, 30, 31, 32, 58, 62, 92, and 93

The sofosbuvir resistance-associated substitution S282T emerged in only 1 non-SVR12 patient infected with genotype 3.

No data are available on the persistence of daclatasvir resistance-associated substitutions beyond 6 months post-treatment in patients treated with daclatasvir and sofosbuvir with/without ribavirin.

Emergent daclatasvir resistance-associated substitutions have been shown to persist for 2 years post- treatment and beyond for patients treated with other daclatasvir-based regimens.

Daclatasvir in combination with peginterferon alfa and ribavirin

Baseline NS5A RAVs were seen in approximately 10% of treatment-naïve patients with genotype 1 infection (GT1a: 28, 30, 31, or 93; GT1b: 31 or 93) and 61% with genotype 4 infection (28, 30, 31). The majority of patients (5/9 [56%] genotype 1a, 6/8 [75%] genotype 1b and 52/57 [91%] genotype 4 patients) with these pretreatment NS5A RAVs described above achieved SVR.

NS5A RAVs generally emerged at failure (139/153 genotype 1a and 49/57 genotype 1b). The most frequently detected NS5A RAVs included Q30E or Q30R in combination with L31M. The majority of genotype 1a failures had emergent NS5A variants detected at Q30 (127/139 [91%]), and the majority of genotype 1b failures had emergent NS5A variants detected at L31 (37/49 [76%]) and/or Y93H (34/49 [69%]).



Hetero

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Pharmacokinetics

The pharmacokinetic properties of daclatasvir in patients with genotype 1 chronic HCV is geometric mean (CV%) daclatasvir C_{max} was 1534 (58) ng/ml, AUC_{0-24h} was 14122 (70) ng•h/ml, and C_{min} was 232 (83) ng/ml

Absorption

Daclatasvir administered as a tablet is readily absorbed following multiple oral doses with peak plasma concentrations occurring between 1 and 2 hours. Daclatasvir C_{max} , AUC, and C_{min} increased in a near dose-proportional manner. Steady state will be achieved after 4 days of once-daily administration. At the 60 mg dose, exposure to daclatasvir is similar between healthy subjects and HCV-infected patients.

The absolute bioavailability of the tablet formulation is 67%.

Effect of food on oral absorption

Administration of daclatasvir 60 mg tablet after a high-fat meal decreased daclatasvir C_{max} and AUC by 28% and 23%, respectively, compared with administration under fasting conditions. Administration of daclatasvir 60 mg tablet after a light meal resulted in no reduction in daclatasvir exposure.

Distribution

At steady state, protein binding of daclatasvir in HCV-infected patients was approximately 99% and independent of dose at the dose range (1 mg to 100 mg). Estimated volume of distribution at steady state was 47 l. Daclatasvir is actively and passively transported into hepatocytes. The active transport is mediated by OCT1 and other unidentified uptake transporters, but not by organic anion transporter (OAT) 2, sodium-taurocholate co transporting polypeptide (NTCP), or OATPs. Daclatasvir is an inhibitor of P-gp, OATP 1B1 and BCRP. Daclatasvir is an inhibitor of renal uptake transporters, OAT1 and 3, and OCT2, but is not expected to have a clinical effect on the pharmacokinetics of substrates of these transporters.

Biotransformation

Daclatasvir is a substrate of CYP3A, with CYP3A4 being the major CYP isoform responsible for the metabolism. No metabolites circulated at levels more than 5% of the parent concentration. Daclatasvir did not inhibit ($IC_{50} > 40 \mu M$) CYP enzymes 1A2, 2B6, 2C8, 2C9, 2C19, or 2D6.

Elimination

88% of Daclatasvir was recovered in feces (53% as unchanged drug) and 6.6% was excreted in the urine (primarily as unchanged drug). Liver is the major clearance organ for daclatasvir in humans. The terminal elimination half-life of daclatasvir ranged from 12 to 15 hours. Total clearance was 4.24 l/h.

Special populations

Renal impairment

Daclatasvir unbound AUC was estimated to be 18%, 39% and 51% higher with creatinine clearance (CL_{cr}) values of 60, 30 and 15 ml/min, respectively, relative to normal renal function. Patients undergoing haemodialysis had a 27% increase in daclatasvir AUC and a 20% increase in unbound AUC compared to patients with normal renal function.

Hepatic impairment

The C_{max} and AUC of total daclatasvir (free and protein-bound drug) were lower in hepatic impairment; however, hepatic impairment did not have a clinically significant effect on the free drug concentrations of daclatasvir.

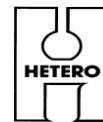
Elderly

Age had no apparent effect on the pharmacokinetics of daclatasvir.

Paediatric population

The pharmacokinetics of daclatasvir in paediatric patients have not been evaluated.

Gender



Hetero

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Female has slightly lower apparent oral clearance CL/F, but the magnitude of the effect on daclatasvir exposure is not clinically important.

Race

Daclatasvir apparent oral clearance (CL/F) and apparent volume of distribution (Vc/F) in patients ([who are not white, black or Asian] and “black”) has slightly higher exposures compared to white patients, but the magnitude of the effect on daclatasvir exposure is not clinically important.

Indications

Dasvir is indicated in combination with other medicinal products for the treatment of chronic hepatitis C virus (HCV) infection in adults (see sections on Recommended Dose, Warnings and Precautions and Pharmacodynamics).

For HCV genotype specific activity, see sections on Warnings and Precautions and Pharmacodynamics

Recommended dose

Treatment with Daclatasvir Dihydrochloride Tablets should be initiated and monitored by a physician experienced in the management of chronic hepatitis C.

Posology

The recommended dose of Daclatasvir Dihydrochloride Tablets is 60 mg once daily, to be taken orally with or without meals.

Daclatasvir Dihydrochloride Tablets must be administered in combination with other medicinal products. The Summary of Product Characteristics for the other medicinal products in the regimen should also be consulted before initiation of therapy with Daclatasvir Dihydrochloride Tablets.

Table 10 provides the recommended Daclatasvir Dihydrochloride Tablets containing treatment regimens and duration based on HCV genotype and patient population. For specific dosage recommendations for sofosbuvir, refer to the prescribing information.

The optimal duration of Daclatasvir Dihydrochloride Tablet and Sofosbuvir with or without ribavirin has not been established for HCV genotype 3 patients with cirrhosis or for HCV genotype 1 patients with Child-Pugh C cirrhosis.

Table 10: Recommended treatment for Daclatasvir Dihydrochloride Tablets interferon - free combination therapy

Patient population	Regimen and duration
HCV Genotype 1	
Without cirrhosis	Daclatasvir Dihydrochloride Tablets + sofosbuvir for 12 weeks
Child Pugh A cirrhosis Child Pugh B or Child Pugh C cirrhosis Post-transplant	Daclatasvir Dihydrochloride Tablets + sofosbuvir + ribavirin for 12 weeks
HCV Genotype 3	
Without cirrhosis	Daclatasvir Dihydrochloride Tablets + sofosbuvir for 12 weeks
Child Pugh A or Child Pugh B or Child Pugh C cirrhosis Post-Transplant	Daclatasvir Dihydrochloride Tablets + sofosbuvir + ribavirin for 12 weeks

* Includes patients co-infected with human immunodeficiency virus (HIV). For dosing recommendations with HIV antiviral agents, refer to section Interaction with other Medicinal Products.

Daclatasvir Dihydrochloride Tablets + peginterferon alfa + ribavirin

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The recommended regimen for patients with genotype 4 infection, without cirrhosis or with compensated cirrhosis is Daclatasvir Dihydrochloride Tablets given for 24 weeks, in combination with 24-48 weeks of peginterferon alfa and ribavirin:

- If HCV RNA is undetectable at both treatment weeks 4 and 12, all 3 components of the regimen should be continued for a total duration of 24 weeks.
- If undetectable HCV RNA is achieved, but not at both treatment weeks 4 and 12, Daclatasvir Dihydrochloride Tablets should be discontinued at 24 weeks and peginterferon alfa and ribavirin continued for a total duration of 48 weeks.

Ribavirin Dosing Guidelines

The dose of ribavirin, when combined with Daclatasvir Dihydrochloride Tablets, is weight-based (1,000 or 1,200 mg in patients <75 kg or ≥75 kg, respectively). Refer to the Summary of Product Characteristics of ribavirin.

For patients with Child-Pugh A, B, or C cirrhosis or recurrence of HCV infection after liver transplantation, the recommended initial dose of ribavirin is 600 mg daily with food. If the starting dose is well-tolerated, the dose can be titrated up to a maximum of 1,000-1,200 mg daily (breakpoint 75 kg). If the starting dose is not well-tolerated, the dose should be reduced as clinically indicated, based on haemoglobin and creatinine clearance measurements (see Table 11).

Table 11: Ribavirin dosing guidelines for coadministration with Daclatasvir Dihydrochloride Tablets regimen for patients with cirrhosis or post transplant

Laboratory Value/Clinical Criteria	Ribavirin Dosing Guideline
Haemoglobin	
>12 g/dL	600 mg daily
> 10 to ≤12 g/dL	400 mg daily
> 8.5 to ≤10 g/dL	200 mg daily
≤8.5 g/dL	Discontinue ribavirin
Creatinine Clearance	
>50 mL/min	Follow guidelines above for haemoglobin
>30 to ≤50 mL/min	200 mg every other day
≤30 mL/min or haemodialysis	Discontinue ribavirin

Dose modification, interruption and discontinuation

Dose modification of Daclatasvir Dihydrochloride Tablets to manage adverse reactions is not recommended. If treatment interruption of components in the regimen is necessary because of adverse reactions, Daclatasvir Dihydrochloride Tablets must not be given as monotherapy. There are no virologic treatment stopping rules that apply to the combination of Daclatasvir Dihydrochloride Tablets with sofosbuvir.

Treatment discontinuation in patients with inadequate on-treatment virologic response during treatment with Daclatasvir Dihydrochloride Tablets, peginterferon alfa and ribavirin

It is unlikely that patients with inadequate on-treatment virologic response will achieve a sustained virologic response (SVR); therefore discontinuation of treatment is recommended in these patients. The HCV RNA thresholds that trigger discontinuation of treatment (i.e. treatment stopping rules) are presented in Table 12.

Table 12: Treatment stopping rules in patients receiving Daclatasvir Dihydrochloride Tablets in combination with peginterferon alfa and ribavirin with inadequate on treatment virologic response

HCV RNA	Action
Treatment week 4: >1000 IU/ml	Discontinue Daclatasvir Dihydrochloride Tablets , peginterferon alfa and ribavirin
Treatment week 12: ≥25 IU/ml	Discontinue Daclatasvir Dihydrochloride Tablets , peginterferon alfa and ribavirin
Treatment week 24: ≥25 IU/ml	Discontinue peginterferon alfa and ribavirin (treatment with Daclatasvir Dihydrochloride Tablets is complete at week 24)

Dose recommendation for concomitant medicines



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Strong inhibitors of cytochrome P450 enzyme 3A4 (CYP3A4)

The dose of Daclatasvir Dihydrochloride Tablets should be reduced to 30 mg once daily when coadministered with strong inhibitors of CYP3A4.

Moderate inducers of CYP3A4

The dose of Daclatasvir Dihydrochloride Tablets should be increased to 90 mg once daily when coadministered with moderate inducers of CYP3A4.

Missed doses

Patients should be instructed that, if they miss a dose of Daclatasvir Dihydrochloride Tablets, the dose should be taken as soon as possible if remembered within 20 hours of the scheduled dose time. However, if the missed dose is remembered more than 20 hours after the scheduled dose, the dose should be skipped and the next dose taken at the appropriate time.

Special populations

Elderly

No dose adjustment of Daclatasvir Dihydrochloride Tablets is required for patients aged ≥ 65 years.

Renal impairment

No dose adjustment of Daclatasvir Dihydrochloride Tablets is required for patients with any degree of renal impairment.

Hepatic impairment

No dose adjustment of Daclatasvir Dihydrochloride Tablets is required for patients with mild (Child-Pugh A, score 5-6), moderate (Child-Pugh B, score 7-9) or severe (Child-Pugh C, score ≥ 10) hepatic impairment.

Paediatric population

The safety and efficacy of Daclatasvir Dihydrochloride Tablets in children and adolescents aged below 18 years have not yet been established. No data are available.

Mode of administration

Daclatasvir Dihydrochloride Tablets are to be taken orally with or without meals. Patients should be instructed to swallow the tablet whole. The film-coated tablet should not be chewed or crushed due to the unpleasant taste of the active substance.

Contraindications

Hypersensitivity to the active substance or to any of the excipients

Coadministration with medicinal products that strongly induce cytochrome P450 3A4 (CYP3A4) and P-glycoprotein transporter (P-gp) and thus may lead to lower exposure and loss of efficacy of Daclatasvir. These active substances include but are not limited to phenytoin, carbamazepine, oxcarbazepine, phenobarbital, rifampicin, rifabutin, rifapentine, systemic dexamethasone, and the herbal product St John's wort (*Hypericum perforatum*).

Warnings and precautions

Dasvir must not be administered as monotherapy. Dasvir must be administered in combination with other medicinal products for the treatment of chronic HCV infection (see sections Indication and Recommended Dose).

Warnings and precautions for other agents in the regimen also apply when co-administered with Dasvir.

Severe bradycardia and heart block

Cases of severe bradycardia and heart block have been observed when Dasvir is used in combination with sofosbuvir and concomitant amiodarone with or without other drugs that lower heart rate. The mechanism is not established.

The concomitant use of amiodarone was limited through the clinical development of sofosbuvir plus direct-acting antivirals (DAAs). Cases are potentially life threatening, therefore amiodarone should only be used in patients on Dasvir and sofosbuvir when other alternative antiarrhythmic treatments are not tolerated or are contraindicated.

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Should concomitant use of amiodarone be considered necessary it is recommended that patients are closely monitored when initiating Dasvirin combination with sofosbuvir. Patients who are identified as being at high risk of bradyarrhythmia should be continuously monitored for 48 hours in an appropriate clinical setting.

Due to the long half-life of amiodarone, appropriate monitoring should also be carried out for patients who have discontinued amiodarone within the past few months and are to be initiated on Dasvir in combination with sofosbuvir.

All patients receiving Dasvir and sofosbuvir in combination with amiodarone with or without other drugs that lower heart rate should also be warned of the symptoms of bradycardia and heart block and should be advised to seek medical advice urgently should they experience them.

Genotype-specific activity

Concerning recommended regimens with different HCV genotypes, see section Recommended Dose. Concerning genotype-specific virological and clinical activity, see section Pharmacodynamic.

Data to support the treatment of genotype 2 infection with Dasvir and sofosbuvir are limited.

Data from study ALLY-3 (AI444218) support a 12-week treatment duration of Dasvir + sofosbuvir for treatment-naïve and -experienced patients with genotype 3 infection without cirrhosis. Lower rates of SVR were observed for patients with cirrhosis (see section Pharmacodynamic).

The clinical data to support the use of Dasvir and sofosbuvir in patients infected with HCV genotypes 4 and 6 are limited. There are no clinical data in patients with genotype 5 (see section Pharmacodynamic).

Patients with Child-Pugh C liver disease

The safety and efficacy of Dasvir in the treatment of HCV infection in patients with Child-Pugh C liver disease have been established in the clinical study ALLY-1 (AI444215, Dasvir + sofosbuvir + ribavirin for 12 weeks); however, SVR rates were lower than in patients with Child-Pugh A and B. Ribavirin may be added based on clinical assessment of an individual patient.

HCV/HBV (hepatitis B virus) co-infection

Cases of hepatitis B virus (HBV) reactivation, some of them fatal, have been reported during or after treatment with direct-acting antiviral agents. HBV screening should be performed in all patients before initiation of treatment. HBV/HCV co-infected patients are at risk of HBV reactivation, and should therefore be monitored and managed according to current clinical guidelines.

Retreatment with daclatasvir

The efficacy of Dasvir as part of a retreatment regimen in patients with prior exposure to a NS5A inhibitor has not been established.

Pregnancy and contraception requirements

Dasvir should not be used during pregnancy or in women of childbearing potential not using contraception. Use of highly effective contraception should be continued for 5 weeks after completion of Dasvir therapy (see section Pregnancy & Lactation).

When Dasvir is used in combination with ribavirin, the contraindications and warnings for that medicinal product are applicable. Significant teratogenic and/or embryocidal effects have been demonstrated in all animal species exposed to ribavirin; therefore, extreme care must be taken to avoid pregnancy in female patients and in female partners of male patients (see the Product Information for ribavirin).

Interactions with medicinal products

Coadministration of Dasvir can alter the concentration of other medicinal products and other medicinal products may alter the concentration of daclatasvir. Refer to section Contraindication for a listing of medicinal products that are

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contraindicated for use with Dasvir due to potential loss of therapeutic effect. Refer to section Interaction with other medicaments for established and other potentially significant drug-drug interactions.

Paediatric population

Dasvir is not recommended for use in children and adolescents aged below 18 years because the safety and efficacy have not been established in this population.

Use in diabetic patients

Diabetics may experience improved glucose control, potentially resulting in symptomatic hypoglycaemia, after initiating HCV DAA treatment. Glucose levels of diabetic patients initiating DAA therapy should be closely monitored, particularly within the first 3 months, and their diabetic medication modified when necessary. The physician in charge of the diabetic care of the patient should be informed when DAA therapy is initiated.

Important information about some of the ingredients in Dasvir

Dasvir contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Interaction with other MedicamentsContraindications of concomitant use

Daclatasvir is contraindicated in combination with medicinal products that strongly induce CYP3A4 and P-gp, e.g. phenytoin, carbamazepine, oxcarbazepine, phenobarbital, rifampicin, rifabutin, rifapentine, systemic dexamethasone, and the herbal product St John's wort (*Hypericum perforatum*), and thus may lead to lower exposure and loss of efficacy of Daclatasvir.

Potential for interaction with other medicinal products

Daclatasvir is a substrate of CYP3A4, P-gp and organic cation transporter (OCT) 1. Strong or moderate inducers of CYP3A4 and P-gp may decrease the plasma levels and therapeutic effect of daclatasvir. Coadministration with strong inducers of CYP3A4 and P-gp is contraindicated while dose adjustment of Daclatasvir is recommended when coadministered with moderate inducers of CYP3A4 and P-gp. Strong inhibitors of CYP3A4 may increase the plasma levels of daclatasvir. Dose adjustment of Daclatasvir is recommended when coadministered with strong inhibitors of CYP3A4. Coadministration of medicines that inhibit P-gp or OCT1 activity is likely to have a limited effect on daclatasvir exposure.

Daclatasvir is an inhibitor of P-gp, organic anion transporting polypeptide (OATP) 1B1, OCT1 and breast cancer resistance protein (BCRP). Administration of Daclatasvir may increase systemic exposure to medicinal products that are substrates of P-gp, OATP 1B1, OCT1 or BCRP, which could increase or prolong their therapeutic effect and adverse reactions. Caution should be used if the medicinal product has a narrow therapeutic range.

Daclatasvir is a very weak inducer of CYP3A4 and caused a decrease in midazolam exposure. However, as this is a limited effect, dose adjustment of concomitantly administered CYP3A4 substrates is not necessary.

Refer to the respective Summary of Product Characteristics for drug interaction information for other medicinal products in the regimen.

Patients treated with vitamin K antagonists

As liver function may change during treatment with Daclatasvir, a close monitoring of International Normalized Ratio (INR) values is recommended.

Tabulated summary of interactions

Table 13 provides information from drug interaction studies with daclatasvir including clinical recommendations for established or potentially significant drug interactions. Clinically relevant increase in concentration is indicated as “↑”, clinically relevant decrease as “↓”, no clinically relevant change as “↔”. The table is not all-inclusive.

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Table 13: Established and Other Potentially Significant Drug Interactions

Concomitant Drug Class: Drug Name	Effect on Concentration	Clinical Comment
ANTIVIRALS, HCV		
<i>Nucleotide analogue polymerase inhibitor</i>		
Sofosbuvir	↔ Daclatasvir ↔ GS-331007 (major circulating metabolite of the prodrug sofosbuvir)	No dose adjustment of Daclatasvir Tablets or sofosbuvir is required.
<i>Protease inhibitors (PIs)</i>		
Boceprevir	↑ Daclatasvir	The dose of Daclatasvir Tablets should be reduced to 30 mg once daily when coadministered with boceprevir or other strong inhibitors of CYP3A4.
Simeprevir	↑ Daclatasvir ↑ Simeprevir	No dose adjustment of Daclatasvir Tablets or sofosbuvir is required.
Telaprevir	↑ Daclatasvir ↔ Telaprevir	The dose of Daclatasvir Tablets should be reduced to 30 mg once daily when coadministered with Telaprevir or other strong inhibitors of CYP3A4.
Other HCV Antivirals		
Peginterferon alfa and ribavirin	↔ Daclatasvir ↔ Ribavirin ↔ Peginterferon alfa	No dose adjustment of Daclatasvir Tablets, Peginterferon alfa or Ribavirin is required.
ANTIVIRALS, HIV OR HBV		
<i>Protease inhibitors (PIs)</i>		
Atazanavir/ritonavir	↑ Daclatasvir	Decrease Daclatasvir Tablet dose to 30mg once daily when coadministered with atazanavir/ritonavir, atazanavir/cobicistat or other strong inhibitors of CYP3A4.
Atazanavir/ cobicistat	↑ Daclatasvir	
Darunavir/ritonavir	↔ Daclatasvir ↔ Darunavir	No dose adjustment of Daclatasvir Tablets, darunavir/ritonavir or Darunavir/cobicistat is required.
Darunavir/cobicistat	↔ Daclatasvir	
Lopinavir/ritonavir	↔ Daclatasvir ↔ Lopinavir	No dose adjustment of Daclatasvir Tablets or Lopinavir/ritonavir is required.
<i>Nucleoside/nucleotide reverse transcriptase inhibitors (NRTIs)</i>		
Tenofovir disoproxil fumarate	↔ Daclatasvir ↔ Tenofovir	No dose adjustment of Daclatasvir Tablets or Tenofovir is required.

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Lamivudine Zidovudine Emtricitabine Abacavir Didanosine Stavudine	↔ Daclatasvir ↔ NRTI	No dose adjustment of Daclatasvir Tablets or the NRTI is required.
<i>Non-nucleoside reverse transcriptase inhibitors (NNRTIs)</i>		
Efavirenz	↓ Daclatasvir	Increase Daclatasvir tablets to 90mg once daily when coadministered with efavirenz
Etravirine Nevirapine	↓ Daclatasvir	Due to the lack of data, coadministration of Daclatasvir Tablets and etravirine or nevirapine is not recommended.
Rilpivirine	↔ Daclatasvir ↔ Rilpivirine	No dose adjustment of Daclatasvir Tablets or rilpivirine is required
<i>Integrase Inhibitors</i>		
Dolutegravir	↑ Dolutegravir ↔ Daclatasvir	No dose adjustment of Daclatasvir Tablets or Dolutegravir is required
Raltegravir	↔ Daclatasvir ↔ Raltegravir	No dose adjustment of Daclatasvir Tablets or Raltegravir is required
Elvitegravir, cobicistat, emtricitabine, tenofovir disoproxil fumarate	↑ Daclatasvir	The dose of Daclatasvir Tablets should be reduced to 30 mg once daily when coadministered with cobicistat or other strong inhibitors of CYP3A4.
<i>Fusion Inhibitors</i>		
Enfuvirtide	↔ Daclatasvir ↔ Enfuvirtide	No dose adjustment of Daclatasvir Tablets or Enfuvirtide is required
<i>CCR5 receptor antagonist</i>		
Maraviroc	↔ Daclatasvir ↔ Maraviroc	No dose adjustment of Daclatasvir Tablets or Maraviroc is required
ACID REDUCING AGENTS		
<i>H₂-receptor antagonists</i>		
Famotidine	↔ Daclatasvir	No dose adjustment of Daclatasvir Tablets is required
<i>Proton Pump Inhibitors</i>		
Omeprazole	↔ Daclatasvir	No dose adjustment of Daclatasvir Tablets is required
ANTIBACTERIALS		
Clarithromycin	↑ Daclatasvir	The dose of Daclatasvir

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Telithromycin		Tablets should be reduced to 30 mg once daily when coadministered with clarithromycin, telithromycin or other strong inhibitors of CYP3A4.
Erythromycin	↑ Daclatasvir	Administration of Daclatasvir tablets with erythromycin may result in increased concentrations of daclatasvir. Caution is advised.
Azithromycin Ciprofloxacin	↔ Daclatasvir ↔ Azithromycin or Ciprofloxacin	No dose adjustment of Daclatasvir tablets or azithromycin or ciprofloxacin is required.
ANTICOAGULANTS		
Dabigatran etexilate	↑ Dabigatran etexilate	Safety monitoring is advised when initiating treatment with Daclatasvir tablets in patients receiving dabigatran etexilate or other intestinal P-gp substrates that have a narrow therapeutic range.
Warfarin or other vitamin K antagonists	↔ Daclatasvir ↔ Warfarin	No dose adjustment of Daclatasvir Tablets or warfarin is required. Close monitoring of INR values is recommended with all vitamin K antagonists. This is due to liver function that may change during treatment with Daclatasvir Tablets
ANTICONVULSANTS		
Carbamazepine Oxcarbazepine Phenobarbital Phenytoin	↓ Daclatasvir	Coadministration of Daclatasvir Tablets with carbamazepine, oxcarbazepine, phenobarbital, phenytoin or other strong inducers of CYP3A4 is contraindicated (see section Contraindications).
ANTIDEPRESSANTS		
<i>Selective serotonin reuptake inhibitors</i>		
Escitalopram	↔ Daclatasvir ↔ Escitalopram	No dose adjustment of Daclatasvir Tablets or Escitalopram is required
ANTIFUNGALS		
Ketoconazole	↑ Daclatasvir	The dose of Daclatasvir



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Itraconazole Posaconazole Voriconazole	↑ Daclatasvir	Tablets should be reduced to 30 mg once daily when coadministered with ketoconazole or other strong inhibitors of CYP3A4.
Fluconazole	↑ Daclatasvir ↔ Fluconazole	Modest increases in concentrations of daclatasvir are expected, but no dose adjustment of Daclatasvir Tablets or fluconazole is required.
ANTIMYCOBACTERIALS		
Rifampicin or Rifabutin or Rifapentine	↓ Daclatasvir	Coadministration of Daclatasvir Tablets with rifampicin, rifabutin, rifapentine or other strong inducers of CYP3A4 is contraindicated (see section Contraindication).
CARDIOVASCULAR AGENTS		
<i>Antiarrhythmics</i>		
Digoxin	↑ Digoxin	Digoxin should be used with caution when coadministered with Daclatasvir Tablets. The lowest dose of digoxin should be initially prescribed. The serum digoxin concentrations should be monitored and used for titration of digoxin dose to obtain the desired clinical effect.
Amiodarone	No Data	Use only if no other alternative is available. Close monitoring is recommended if this medicinal product is administered with Daclatasvir Tablets in combination with sofosbuvir (see sections Warning & Precautions and Side Effect).
<i>Calcium channel blockers</i>		
Diltiazem Nifedipine Amlodipine Verapamil	↑ Daclatasvir	Administration of Daclatasvir Tablets with any of these calcium channel blockers may result in increased concentrations of daclatasvir. Caution is advised.
CORTICOSTEROIDS		

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Systemic dexamethasone	↓ Daclatasvir	Coadministration of Daclatasvir Tablets with systemic dexamethasone or other strong inducers of CYP3A4 is contraindicated (see section Contraindications).
HERBAL SUPPLEMENTS		
St. John’s wort (<i>Hypericum perforatum</i>)	↓ Daclatasvir	Coadministration of Daclatasvir Tablets with St. John’s wort or other strong inducers of CYP3A4 is contraindicated (see section Contraindications).
HORMONAL CONTRACEPTIVES		
Ethinylestradiol + norgestimate	↔ Ethinylestradiol ↔ Norelgestromin ↔ Norgestrel	An oral contraceptive containing ethinylestradiol and norgestimate is recommended with Daclatasvir Tablets. Other oral contraceptives have not been studied.
IMMUNOSUPPRESSANTS		
Cyclosporine	↔ Daclatasvir ↔ Cyclosporine	No dose adjustment of either medicinal product is required when Daclatasvir Tablets is coadministered with cyclosporine, tacrolimus, sirolimus or mycophenolate mofetil.
Tacrolimus	↔ Daclatasvir ↔ Tacrolimus	
Sirolimus Mycophenolate mofetil	↔ Daclatasvir ↔ Immunosuppresant	
LIPID LOWERING AGENTS		
HMG-CoA reductase inhibitors		
Rosuvastatin	↑ Rosuvastatin	Caution should be used when Daclatasvir Tablets is coadministered with rosuvastatin or other substrates of OATP 1B1 or BCRP.
Atorvastatin Fluvastatin Simvastatin Pitavastatin Pravastatin	↑ Concentration of statin	
NARCOTIC ANALGESICS		
Buprenorphine/naloxone	↔ Daclatasvir ↑ Buprenorphine ↑ Norbuprenorphine	No dose adjustment of Daclatasvir Tablets or buprenorphine may be required, but it is recommended that patients should be monitored for signs of opiate toxicity.
Methadone	↔ Daclatasvir ↔ R-methadone	No dose adjustment of Daclatasvir Tablets or Methadone is required
SEDATIVES		
Benzodiazepines		
Midazolam	↔ Midazolam	No dose adjustment of



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Triazolam Alprazolam	↔ Triazolam ↔ Alprazolam	midazolam, other benzodiazepines or other CYP3A4 substrates is required when coadministered with Daclatasvir Tablets
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No clinically relevant effects on the pharmacokinetics of either medicinal product are expected when daclatasvir is coadministered with any of the following: PDE-5 inhibitors, medicinal products in the ACE inhibitor class (e.g. enalapril), medicinal products in the angiotensin II receptor antagonist class (e.g. losartan, irbesartan, olmesartan, candesartan, valsartan), disopyramide, propafenone, flecainide, mexilitine, quinidine or antacids.

Pregnancy and lactation

Pregnancy

There are no data from the use of daclatasvir in pregnant women.

Animal studies have shown embryotoxic and teratogenic effects. The potential risk for humans is unknown.

Daclatasvir should not be used during pregnancy or in women of childbearing potential not using contraception. Use of highly effective contraception should be continued for 5 weeks after completion of daclatasvir therapy.

Since daclatasvir is used in combination with other agents, the contraindications and warnings for those medicinal products are applicable.

For detailed recommendations regarding pregnancy and contraception, refer to the Summary of Product Characteristics for ribavirin and peginterferon alfa.

Breast-feeding

It is not known whether daclatasvir is excreted in human milk. Available pharmacokinetic and toxicological data in animals have shown excretion of daclatasvir and metabolites in milk (. A risk to the newborn/infant cannot be excluded. Mothers should be instructed not to breastfeed if they are taking Daclatasvir.

Fertility

No human data on the effect of daclatasvir on fertility are available.

Undesirable Effects

Summary of the safety profile

Daclatasvir in combination with sofosbuvir

The most frequently reported adverse reactions were fatigue, headache, and nausea.

Daclatasvir in combination with peginterferon alfa and ribavirin

The most frequently reported adverse reactions were fatigue, headache, pruritus, anaemia, influenza-like illness, nausea, insomnia, neutropenia, asthenia, rash, decreased appetite, dry skin, alopecia, pyrexia, myalgia, irritability, cough, diarrhoea, dyspnoea and arthralgia. The most frequently reported adverse reactions of at least Grade 3 severity (frequency of 1% or greater) were neutropenia, anaemia lymphopenia and thrombocytopenia. The safety profile of daclatasvir in combination with peginterferon alfa and ribavirin was similar to that seen with peginterferon alfa and ribavirin alone, including among patients with cirrhosis.

Tabulated list of adverse reactions

Adverse reactions are listed in Table 14 by regimen, system organ class and frequency: Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

**Hetero****DASVIR 30/60
(DACLATASVIR TABLETS 30 mg & 60 mg)****Table 14: Adverse reactions**

System Organ Class	Adverse Reactions	
Frequency	<i>Dasvir +sofosbuvir + ribavirin</i>	<i>Dasvir +sofosbuvir</i>
Blood and lymphatic system disorders		
very common	anaemia	
Metabolism and nutrition disorders		
common	decreased appetite	
Psychiatric disorders		
common	insomnia, irritability	insomnia
Nervous system disorders		
very common	headache	headache
common	dizziness, migraine	dizziness, migraine
Vascular disorders		
common	hot flush	
Respiratory, thoracic and mediastinal disorders		
common	dyspnoea, dyspnoea exertional, cough, nasal congestion	
Gastrointestinal disorders		
very common	nausea	
common	diarrhoea, vomiting, abdominal pain, gastrooesophageal reflux disease, constipation, dry mouth, flatulence	nausea, diarrhoea, abdominal pain
Skin and subcutaneous tissue disorders		
common	rash, alopecia, pruritus, dry skin	
Musculoskeletal and connective tissue disorders		
common	arthralgia, myalgia	arthralgia, myalgia
General disorders and administration site conditions		
very common	fatigue	fatigue

Laboratory abnormalities

Daclatasvir Dihydrochloride Tablets in combination with sofosbuvir with or without ribavirin, 2% of patients had Grade 3 haemoglobin decreases; all of these patients received Daclatasvir Dihydrochloride Tablets + sofosbuvir + ribavirin. Grade 3/4 increases in total bilirubin were observed in 5% of patients (all in patients with HIV co-infection who were receiving concomitant atazanavir, with Child-Pugh A, B, or C cirrhosis, or who were post-liver transplant).

Description of selected adverse reactions**Cardiac arrhythmias**

Cases of severe bradycardia and heart block have been observed when Daclatasvir Dihydrochloride Tablets are used in combination with sofosbuvir and concomitant amiodarone and/or other drugs that lower heart rate (see sections Warning and Precautions and Interactions with Other Medicaments).

Paediatric population

The safety and efficacy of Daclatasvir Dihydrochloride Tablets in children and adolescents aged <18 years have not yet been established. No data are available.



Hetero

DASVIR 30/60 (DACLATASVIR TABLETS 30 mg & 60 mg)

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product.

Symptoms and treatment of overdose

There is no known antidote for overdose of daclatasvir. Treatment of overdose with daclatasvir should consist of general supportive measures, including monitoring of vital signs, and observation of the patient's clinical status. Because daclatasvir is highly protein bound (99%) and has a molecular weight >500, dialysis is unlikely to significantly reduce plasma concentrations of daclatasvir.

Effects on Ability to Drive and Use Machine

Dizziness has been reported during treatment with daclatasvir in combination with sofosbuvir, and dizziness, disturbance in attention, blurred vision and reduced visual acuity have been reported during treatment with daclatasvir in combination with peginterferon alfa and ribavirin.

Storage condition Store below 30°C and protect from moisture.

Shelf life: 2 Years

Container pack: 28's tablets in a HDPE bottle

Manufacturer: HETERO LABS LIMITED, UNIT-V, Sy. No 439, 440, 441 & 458, TSIIC Formulation SEZ, Polepally Village, Jadcherla Mandal, Mahaboobnagar Dist., Telangana, India.

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Date of Revision: September 2019